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Driving success in closed-loop supply chain: investigating consumer attitudes toward e-waste return

Imnatila Pongen
T A Pai Management Institute, Manipal Academy of Higher Education, Manipal, India and
Indian Institute of Management Ranchi, Ranchi, India
Pritee Ray and T.S. Vijay
Indian Institute of Management Ranchi, Ranchi, India, and
Kannan Govindan
Centre for Sustainable Operations and Resilient Supply Chains, Adelaide Business School, Institute for Sustainability, Energy and Resources, The University of Adelaide, Adelaide, Australia and
Centre for Sustainable Supply Chain Engineering, Department of Technology and Innovation, Danish Institute for Advanced Study, University of Southern Denmark, Sønderborg, Denmark

Abstract

Purpose – Implementation of an effective closed-loop supply chain (CLSC) requires knowledge of consumers' perceptions of returning end-of-life products. Consumers, as primary suppliers of end-of-life products, play a crucial role in recycling efforts. This study identifies the drivers behind the e-waste recycling intentions of consumers, bridging the gap in previous studies.

Design/methodology/approach – Extending the Theory of Planned Behavior, this research integrates other determinants applicable to e-waste return intentions, including eco-literacy, reverse logistics awareness, perceived risk and incentives. An online survey was conducted with 310 digitally literate Indian consumers. Structural equation modeling was used to analyze the relationships among the studied determinants. Fourteen hypotheses were tested using PLS-SEM to assess direct and moderating effects.

Findings – The results show significant relationships between the determinants and return intention. Moderation analysis indicates that incentives reinforce the positive impact of perceived behavioral (PB) control on return intention, while perceived risk undermines the attitude-return intention relationship, as well as the return intention–PB control relationship. Attitude, subjective norms, eco-literacy, collection method and incentives were significant positive predictors, while perceived risk acted as a key barrier.

Practical implications – Our study indicates that creating awareness programs on the importance of e-waste recycling, implementing incentive schemes and establishing more stringent e-waste management regulations can promote returning e-waste into the supply chain.

Originality/value – This study contributes to the literature on CLSCs by conceptualizing consumer e-waste return intention using an extended TPB applied in the Indian context – fast-emerging digitalized economy with a high percentage of informal recycling. Through the integration of cognitive (eco-literacy, reverse logistics awareness), contextual (collection mechanism, perceived risk) and motivational (rewards) determinants, the study empirically tests the moderating roles of rewards and risk through structural equation modeling, providing refined theoretical insights and practical implications for policymakers and industry stakeholders.

Keywords Structural equation modeling, e-waste management, Return intention, Consumer behavior, Closed-loop supply chain

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1. Introduction

The rise in technological advancement and consumer preference for newer electronics contributes significantly to accelerate the rate of device obsolescence. The global generation of

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waste electrical and electronic equipment (WEEE) totaled 62 million metric tons, but only 17.4% was officially collected and recycled. India is one of the world's leading producers of e-waste, with an estimated 5 million metric tons produced in 2024 alone (Global E-Waste Monitor, 2024). The formal collection rate in India is still less than 15%, and the cost of disposing e-waste in an environmentally responsible manner varies from ₹30 to ₹ 80 per kilogram, depending on the type of device and processing method (CPCB, 2023). These statistics emphasize the need for better and more sustainable e-waste management practices.

The constantly shortening life-cycle of electronic products – like computers, laptops and mobile phones – calls for innovation and implementation of effective and quicker return mechanisms. However, e-waste collection and appropriate disposal systems are falling behind compared to the pace at which the products reach End-of-Life (EoL) or End-of-Use (EoU). This disparity is causing serious harm to society and the environment (Khan *et al.*, 2019; Najmi *et al.*, 2021). To mitigate the effects of inappropriate disposal of e-waste, it has become crucial to consider the adoption of closed-loop supply chain (CLSC) through reverse logistics (Khan *et al.*, 2019).

To ground this research in the wider academic discourse, we highlight the critical contextual uniqueness informing our inquiry. India offers a singular confluence of fast-paced digitalization, exponential adoption of consumer electronics and a nascent formal e-waste management ecosystem. This synergy provides a dynamic yet undertheorized research environment, offering grounds for developing and advancing theoretical frameworks for understanding sustainable consumer behavior.

The complexity and urgency of this socio-technical environment require a contextually informed inquiry that can capture the dynamic behavioral patterns involved in e-waste return. By situating our study in this high-growth, high-risk context, we gain theoretically sound, empirically rich and practically relevant insights. In addition, the contextual subtleties identified in this study increase the transferability and adaptability of our conclusions to other rapidly digitalizing economies facing similar sustainability issues. This research thereby meaningfully contributes to both theory-building and practical policy design in the wider field of sustainable supply chain management.

Extant research indicates that some customers are unwilling to take part (Masi *et al.*, 2018), while some reluctantly return the used product after the time limit (Park *et al.*, 2010). These factors add to the uncertainty surrounding EoL recovery (Kumar *et al.*, 2019; Saidani *et al.*, 2019; Wang *et al.*, 2022). Previous studies attribute the low return rate of used products largely to the lack of awareness about collection systems (Geyer and Doctori Blass, 2010; Jena and Sarmah, 2015). Most consumer-related studies in the literature are focused on adopting remanufactured products. In contrast, studies focused on consumers' intentions in the broader context of e-waste management, i.e. return, reuse, recycle and refurbish, are scarce. This research directly addresses this gap by developing a context-specific behavioral model.

Recognizing the importance of consumer participation, this study aims to address the underlying determinants of consumer intention toward e-waste management. To achieve this aim, this study extends the Theory of Planned Behavior (TPB) by integrating constructs relevant to the Indian demography. The purpose of this study is threefold: (1) examine the relationship between key psychological and contextual constructs and the consumers' return intention of e-waste, (2) identify the most influential constructs shaping the consumers' return intention and (3) explore the moderating role of perceived risk on the relationship between return intention and attitude, perceived behavioral (PB) control and awareness consequences, along with the moderating effect of incentives on the relationship between return intention and PB control.

Guided by these objectives, we aim to address the following research questions:

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RQ1. What psychological and contextual determinants shape consumers' return intention for e-waste?

RQ2. What are the most significant determinants shaping return intention?

RQ3. What moderating effects do incentives and perceived risk have on the associations between return intention and important TPB components?

A structured survey of Indian consumers was conducted to address the research questions, and the proposed model was tested using Partial Least Squares Structural Equation Modeling (PLS-SEM).

This study adds substantially to both practical and theoretical insights. Theoretically, it increases the explanatory potential of TPB in the sustainable supply chain by including context-related moderators like incentives and perceived risk. Practically, it provides actionable recommendations to policymakers and stakeholders of electronic devices for improving return infrastructure, promoting purposeful consumer awareness campaigns and creating successful regulatory and incentive-driven frameworks for enabling environmentally conscious behavior. Placed in the fast-changing Indian digital economy, the findings of the study also provide translatable recommendations to other emerging markets facing the same sustainability issues regarding e-waste management.

The paper is organized as follows. The theoretical background and hypothesis are presented in Section 2. Section 3 describes the research methodology, followed by data analysis in Section 4. Section 5 presents the theoretical and managerial implications of the study and concludes the paper in Section 6.

2. Literature review

2.1 Theoretical background

The psychological mechanism behind human action is complex and understanding this human psychology is critical to explaining behavior toward different stimuli. One of the most widely adopted frameworks for modeling such behavior is the TPB. TPB proposes that three key factors – attitude, subjective norms and PB control – collectively determine an individual's behavioral intention (Ajzen, 1991). These constructs shape personal beliefs, perceptions of social influence and perceptions of ease or difficulty for carrying out behavioral actions. Although TPB offers a strong framework, researchers tend to modify it to better fit specific research domains through the addition of more situational variables (Wan *et al*., 2012; Alhassan *et al*., 2018).

E-waste poses special challenges involving data privacy, high recovery costs and the need for specialized logistics infrastructure requirements. Although recycling behavior has been studied in the literature, there is still limited research on e-waste return behavior, focusing on emerging markets. This research fills that gap by situating the TPB in context with pertinent other constructs such as eco-literacy, reverse logistics awareness and perceived risk.

To tackle e-waste, numerous return programs have been established globally. For instance, the European Union's Waste Electrical and Electronic Equipment (WEEE) Directive mandates producers to facilitate product return and recycling (Kumar and Dixit, 2021). In the USA, the Electronic Waste Recycling Act has been enacted in California to encourage recycling (Forti *et al*., 2020). In the developing world, manufacturer-initiated e-waste return systems such as HP's "Planet Partners" and Xiaomi phone exchange program in India have been effective (Zeng *et al*., 2017; Ghosh *et al*., 2022).

Evidence shows that the incorporation of operation and context constructs into TPB enhances its predictability for sustainable behaviors (Adaryani *et al*., 2025; Wang *et al*., 2024).

Past research on electronic waste has focused on either building or improving reverse logistics or measuring consumer behavior using socio-psychological paradigms, with little or no integration of the two fields. This study tries to bridge the gap by applying the TPB,

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incorporating operational variables such as reverse logistics awareness, collection methods and incentives. This proposed framework integrates the behavioral and the logistics aspects and contributes theoretically as well as provides practical suggestions for improving e-waste return systems in developing economies.

Accordingly, the current study builds on this integration by including six operational and informational constructs – reverse logistics awareness, eco-literacy, awareness consequences, collection methods, perceived risk and incentives – that are particularly relevant to emerging markets. Specifically, the model adopted eco-literacy to improve the cognitive basis of attitude and perceived control of behavior, knowledge of reverse logistics and collection method as control facilitators through system familiarity and availability, awareness consequences to improve attitude measurement, perceived risk as an external constraint that can affect the impact of attitude and control on intention and incentives as external motivator to improve return intentions beyond what TPB originally considered.

The selection of constructs for this study is based on evidence from earlier research as well as their applicability to developing economies. Previous studies identified constructs such as moral obligation and past behavior as having an impact on return intention (Nigbur *et al.*, 2010; Zhang *et al.*, 2020); however, to preserve model parsimony and concentrate on informational, infrastructural and motivational variables that are more actionable for policy design in emerging economies, these constructs were eliminated.

By combining these constructs, this study presents a novel extension of the TPB framework to elucidate the behavioral, informational and infrastructural factors influencing e-waste return behavior. This not only improves the model's explanatory capacity but also provides practical implications for devising effective interventions to encourage responsible consumer behavior. Figure 1 illustrates the conceptual model showing the relationships between the studied constructs and return intention.

2.2 Hypotheses development

2.2.1 Attitude. Attitude (AT) refers to a person's learned tendency to evaluate a specific behavior – for example: returning e-waste – either positively or negatively, depending on acquired experience, beliefs and values. It denotes emotional, cognitive and behavioral aspects of the individual's response toward the behavior. In this study, attitude encompasses the degree to which consumers feel that returning used electronics is favourable for the environment, society and themselves.

Studies found that a positive attitude significantly impacts on behaviors such as buying remanufactured goods (Khor and Hazen, 2017), reusing old products (Khan *et al.*, 2019) and

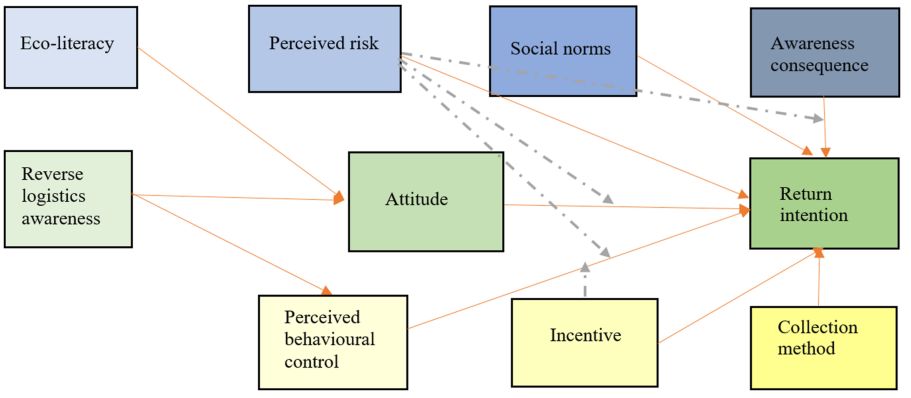


Figure 1. Proposed conceptual model. Source: Authors' own work

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intention to recycle (Vijayan *et al*., 2023; Sabbir *et al*., 2023). Recent studies also emphasized that positive attitudes are important in promoting sustainable behavior (Puzzo and Prati, 2024; Adaryani *et al*., 2025). However, there are some contradictions where attitude had a negligible or no adverse effect (Park and Ha, 2014; Lizin *et al*., 2017). In this study, we define attitude as consumers' perception of e-waste return as a socially responsible act and positive action to society. This aligns with Singh *et al.* (2025), who established that positive emotions and individual beliefs were significant in shaping individuals' intention to shop in the metaverse, corroborating that attitudes play a critical role in sustainable consumer actions. We hypothesize.

H1. Attitude has a positive influence on the e-waste return intention of consumers

2.2.2 Subjective norms. Subjective norms (SN) are the perceived social norms and normative pressures felt by individuals from significant referent groups, e.g., family members, friends, colleagues and community. Subjective norms, in the context of e-waste, are the perceived notion that one's close circle wants them to perform environmentally friendly disposal practices and thereby affect their intention to use return programs (Ajzen, 1991; Si *et al*., 2020).

Empirical studies showed subjective norms to influence recycling activities (Park and Ha, 2014), such as intentions to reuse plastic (Khan *et al*., 2019) and information, communication and technology products (Najmi *et al*., 2021).

Given the strong collectivist tendencies found in emerging economies such as India, social influence can potentially have an even stronger influence. Therefore:

H2. Subjective norm has a positive impact on consumer return intention

2.2.3 Perceived behavioral control. PB control indicates a person's belief of the simplicity or complexity of carrying out a behavior, depending on internal determinants (e.g. self-efficacy) and external circumstances (e.g. access to infrastructure, time availability). In the case of e-waste, it represents consumers' confidence in their capacity to find, access and use formal channels for returning used devices.

Studies concur with the idea that higher perceived control is associated with a higher probability of engaging in recycling behavior (Khan *et al*., 2019; Sari *et al*., 2021; Vijayan *et al*., 2023). Lack of time, knowledge or logistic support often inhibits this sense of control.

Therefore, we hypothesize:

H3. PB control has a positive impact on consumer return intention

2.2.4 Reverse logistics awareness. Reverse logistics (RL) awareness refers to the consumers' knowledge and understanding of the mechanisms, availability and environmental significance of reverse flows of products (e-waste). RL awareness includes knowledge about recycling programs, drop-off locations, company take-back programs and benefits from appropriate disposal. High reverse logistics awareness consumers are most likely to believe that returns of products are easy and worthwhile.

Research indicates that a lack of RL awareness is a significant limitation to effective CLSCs (Bekchanov and Mirzabaev, 2018; Ferronato *et al*., 2019; Yahya *et al*., 2021). It can be inferred that customers with high reverse logistics awareness will most likely feel that product return is easy and worthwhile.

Thus, increased awareness is expected to enhance both attitude and perceived control:

H4. Reverse logistics awareness has a positive influence on the e-waste return attitude of consumers

H5. Reverse logistics awareness has a positive influence on the PB control of consumers

2.2.5 Eco-literacy. Eco-literacy (EL) refers to a person's overall awareness of ecological systems, environmental risks and the influence of human action on the environment.

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Concerning e-waste, EL represents consumers' awareness about resource conservation, pollution hazards and sustainable practices, along with the desire to adopt behaviors for reducing environmental impact.

Studies have also emphasized the importance of eco-literacy as a key facilitator of pro-environmental behavior and the uptake of sustainable practices such as CLSC (Hazen *et al*., 2017; Mathivathanan *et al*., 2022). Recent studies (Pappas *et al*., 2023; Najafabadiha *et al*., 2025) also confirmed the role of EL on shaping pro-environmental intentions. Environmentally conscious consumers make educated decisions and are more likely to behave in environmentally responsible ways. Tripathi and Trigunait (2025) also showed that knowledge of the environment – closely linked to eco-literacy – was a good predictor of sustainable behaviors like recycling and green product use.

Thus:

H6. Eco-literacy has a positive impact on the attitude toward consumers' return intention of e-waste

2.2.6 Collection method. Collection method (CM) captures the design and execution of systems intended to facilitate the return of used products. This includes the variety, accessibility, convenience and reliability of available return channels, whether physical (drop-off centers) or digital (online scheduling for pickups). A well-designed collection method can reduce perceived effort and significantly increase consumer participation rates.

Studies indicate that proper design increases participation (Du and Lang, 2015; Yahya *et al*., 2021). Strategically implemented collection channels minimize transaction costs and increase consumer engagement (Lyu *et al*., 2023; Puzzo and Prati, 2024). Lack of convenient options can discourage even well-intentioned consumers.

Thus:

H7. Availability of collection channel has a positive impact on the consumer return intention toward recycling e-waste

2.2.7 Awareness consequences. Awareness consequences (AC) refers to an individual's recognition and understanding of the broader environmental, social and health-related outcomes of their actions or inactions regarding waste disposal. In e-waste return behavior, this means appreciating the benefits of recycling (such as pollution reduction and resource recovery) and the disadvantages of improper disposal (such as toxic leakage and ecosystem damage).

Previous studies (Wang *et al*., 2016; Lakhan, 2018) suggest that increased awareness consequences tend to motivate recycling behaviors, although a few studies found neutral effects (Afroz *et al*., 2015; Sabbir *et al*., 2023).

Thus, we propose the following hypothesis.

H8. Awareness consequences has a positive impact on the consumer return intention toward recycling e-waste

2.2.8 Perceived risk. PR refers to the consumer's subjective assessment of potential adverse outcomes arising from returning used electronics. Risk factors play a crucial role in determining consumer behavior and are usually modelled as a determining factor of intention through attitude. It mainly includes concerns about data security breaches, unauthorized use of personal data, damage during collection or inaccessibility of reliable recyclers. Fear of privacy infringement during recycling discourages returns (Lyu *et al*., 2023). Perceived risk can, therefore, repress otherwise positive effects like attitude and PB control (Yin *et al*., 2014; Zhang *et al*., 2020).

Thus, we hypothesize:

H9. Perceived risk has a positive impact on the consumer return intention toward recycling e-waste

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H10. If an individual perceives high risk, then the relationship between the person's attitude and e-waste return intention will be weakened

H11. If an individual perceives high risk, then the relationship between the person's PB control and e-waste return intention will be weakened

H12. If an individual perceives high risk, then the relationship between the person's awareness of consequences and e-waste return intention will be weakened

2.2.9 Incentive. Incentive (IN) encompasses both tangible rewards (such as monetary compensation, discounts and rebates) and intangible rewards (such as environmental recognition or social status) provided to motivate or stimulate preferred behavior. For e-waste recycling, incentives act as a motivator by balancing perceived effort, inconvenience or risk involved in returning EoL and EoU products.

Studies have indicated that incentives have a positive effect on consumers' willingness to engage in recycling programs (Yin *et al.*, 2014; Sabbir *et al.*, 2023). Even modest rewards can tip behavioral intention favourably (Liu *et al.*, 2017; Puzzo and Prati, 2024).

Thus, we hypothesize:

H13. Financial incentives have a positive influence on the consumers' return intention toward recycling e-waste

H14. Incentive will positively influence the relationship between the PB control and e-waste return intention

This study adds significantly to the literature on TPB and sustainability behavior in four ways:

(1) It moves the emphasis away from the adoption of environmentally friendly products and toward the less-researched consumer choice to return electronics that have reached the end of their useful lives.

(2) It integrates six new concepts empirically into the TPB framework to make it more contextually relevant to emerging market environments.

(3) It simultaneously models both perceived risk and incentives as moderators – which have rarely been tested simultaneously in previous recycling intention studies.

(4) Collectively, this extended TPB model presents a more holistic framework for understanding and shaping consumer e-waste return behavior in developing economies.

A summary of the definition for the considered constructs is given in Table 1.

3. Research methodology

3.1 Designing the questionnaire

This study employs a quantitative approach to exploring the determinants of consumers' intentions to recycle e-waste. A structured survey questionnaire was designed for this study based on the extended TPB. Measurement indicators for the study were identified from previous studies and fine-tuned into concise, simple statements to fit the understanding capabilities of the target respondent population. The items were measured using a five-point Likert scale of agreeableness from strongly disagree to strongly agree. The sources for the measurement indicators are summarized in Table 2, and the construct items are in Appendix.

To ascertain instrument reliability and validity, a pilot survey of 30 doctoral students was undertaken, following the recommendations by Hertzog (2008). The purpose of the pilot survey was to assess item clarity, establish internal consistency and construct validity. Based on feedback from participant scholars, minor modifications were made to the wording of a few items to improve comprehension and remove ambiguity. Specifically, some selected items under the

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Table 1. A summary of definitions for constructs

Construct	Definition summary
Attitude	A learned tendency to assess returning e-waste as good or bad depending on experience and beliefs
Subjective Norms	Perceived social norms from significant referents that affect the e-waste recycling behavior of the consumer
Perceived Behavioral Control	Consumer's perceived ease or complexity in returning e-waste, depending on resources and situation
Reverse Logistics Awareness	Consumer's awareness of the availability, accessibility and environmental function of e-waste return channels
Eco-literacy	Consumer's awareness of ecological principles, environmental hazards and dedication to sustainable practices
Collection Method	The ease of access, convenience and return channel system allowing consumer involvement
Awareness Consequences	Identification of environmental, social and health consequences arising from recycling or illegal e-waste dumping
Perceived Risk	Subjective assessment of negative consequences such as privacy violations from engaging in e-waste return programs
Incentive	Rewards, including monetary and non-monetary incentives, provided to encourage consumers to return used electronics

Source(s): Authors' own work

Table 2. Sources of measuring items

Construct	References
Attitude	Khan *et al.* (2019), Zhang *et al.* (2020)
Subjective norm	Najmi *et al.* (2021), Zhang *et al.* (2020)
Perceived behavioral control	Khor and Hazen (2017), Zhang *et al.* (2020)
Reverse logistics awareness	Liu *et al.* (2017), Hazen *et al.* (2012)
Eco-literacy	Babaei *et al.* (2015), Kianpour *et al.* (2017)
Collection method	Kianpour *et al.* (2017)
Awareness consequences	Kumar (2019), Wang *et al.* (2021)
Perceived risk	Kumar (2019), Zhang *et al.* (2020)
Incentive	Kianpour *et al.* (2017)
Return intention	Kianpour *et al.* (2017), Kumar (2019), Zhang *et al.* (2020)

Source(s): Authors' own work

"collection method" construct were further refined to help define how multichannel availability, formal recycling infrastructure and system acceptability facilitate e-waste return. Moreover, two items from other constructs were removed because they had low item-total correlations (<0.40). Reliability analysis showed that all constructs had acceptable internal consistency after these changes, with Cronbach's alpha values above the suggested cutoff of 0.70.

3.2 Data collection

A non-probability convenience sampling technique was employed. The questionnaire was distributed online to a larger group of respondents through professional networks, university mailing lists and widely used social media sites such as LinkedIn, WhatsApp and Telegram. The final questionnaire was administered online. Online administration facilitated broader geographical reach and reduced interviewer bias, as well as social desirability tendencies through anonymous response. Participation in the study was entirely voluntary and did not entail any payment.

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The questionnaire has three sections: (1) a basic description of the research to familiarize the respondents with the study area, (2) demographic profiles with questions on gender, age and education level and (3) measurement constructs, including attitude, social norms, PB control, eco-literacy, reverse logistics awareness, incentive, return intention, collection method, awareness consequences and perceived risk. The items for the studied construct can be found Table A1.

A screening question was employed to screen respondent eligibility: "Do you currently live in India and have at least one electronic product?" Only those participants who responded positively were allowed to continue, thus ensuring they were part of the study's digitally literate Indian consumer target. This group represents a key consumer segment that is likely to engage in e-waste return behavior of specific interest to developing digital economies like India. The sample captured significant demographic diversity across gender, age and education, despite the fact that convenience sampling restricts generalizability to the full population. This supports the findings' broader applicability to India's urban, digitally connected population –an important target for sustainable e-waste return policy and reverse logistics infrastructure.

Since a single-source, self-reported survey procedure was used, potential common method variance (CMV) was averted *a priori* via both procedural and statistical methods. Procedurally, the wording of items was made neutral and non-leading and the questions were randomized to reduce the likelihood of response biases as well as randomized item order. Respondents were guaranteed complete anonymity and confidentiality in order to minimize evaluation apprehension.

Post hoc statistical testing for residual CMV was performed. To that end, Harman's Single Factor Test was performed using exploratory factor analysis involving principal component analysis without rotation of factors. Results indicated that the first unrotated factor explained 35.73% of the total variance – far below the 50% mark – meaning that CMV was not a significant issue.

Subsequently, a complete collinearity test was also performed according to the methodology suggested by Kock and Lynn (2012). The variance inflation factors (VIFs) for all the constructs were less than the suggested threshold of 3.33, which validated that there was no excessive multicollinearity and further reduced concerns for CMV.

Therefore, through the complementary combination of preventive design techniques and post hoc testing, this research efficiently reduces the likelihood of common method bias (CMB), thus strengthening the methodological robustness and the validity and reliability of the results.

4. Data analysis

At the end of data collection, we received 310 completed responses. There were no missing values on all the items in the received response. A summary of the demographic profile of the respondents is provided in Table 3. It shows that the sample had 60.96% of male respondents, and 95% of the respondents were millennials and Gen-Z consumers. Of the total respondents, 31.93% were graduates, and 57.74% were postgraduates or above.

Table 4 shows descriptive statistics of the primary latent constructs. The highest mean scores were observed for attitude (M = 4.34) and consequences awareness (M = 4.32), indicating that consumers have positive attitudes toward responsible e-waste behavior. Reverse logistics awareness (M = 3.17) and perceived risk (M = 3.23) had comparatively lower means and greater standard deviations, implying greater variability and indicating the need for specially designed awareness and assurance programs are required.

Based on the objectives of this study, data for the proposed model are analyzed using SmartPLS 4. We adopted PLS-SEM to establish the correlation between the variables and study the constructs' significance. We first conducted outer loading analysis, followed by model measurement, and finally, hypothesis testing of the structural model according to the PLS-SEM guidelines (Hair *et al.*, 2021). We used SmartPLS 4 as it allows for moderation analysis directly.

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Table 3. Descriptive statistics of respondents ($N = 310$)

Demographic		*N*	Percentage (%)
Gender	Male	189	61.0
	Female	119	38.4
	Prefer not to say	2	0.6
Age	Below 20	2	0.6
	21 to 30	247	79.7
	31 to 40	45	14.5
	41 and above	16	5.1
Education	Undergraduate	32	10.3
	Graduate	99	32.0
	Postgraduate and above	179	57.7

Source(s): Authors' own work

Table 4. Descriptive statistics of constructs ($N = 310$)

Construct	Number of items	Mean	Standard deviation
Attitude (AT)	5	4.33742	0.54287
Subjective Norm (SN)	4	3.46532	0.73282
Perceived Behavioral Control (PB)	4	3.5379	0.70019
Reverse Logistics Awareness (RL)	3	3.16774	1.12784
Eco-literacy (EL)	3	4.28925	0.64628
Collection Method (CM)	3	4.05161	0.60519
Awareness Consequences (AC)	4	4.32419	0.56005
Perceived Risk (PR)	3	3.22796	0.99025
Incentives (IN)	3	3.83118	0.7854

Source(s): Authors' own work

4.1 Measurement model analysis

The proposed model was analyzed to evaluate its reliability and validity. An outer loading analysis was carried out to assess the factor loadings of the items in the questionnaire, which are presented in Table 5.

4.1.1 Reliability testing. Reliability testing was carried out to evaluate the internal consistency of the measurement scale (Neuman, 2007). Two indicators were employed: Cronbach's alpha and composite reliability. Cronbach's alpha is used for assessing the internal adjustments, that is, the extent to which one construct is correlated with the items forming the construct. Cronbach's alpha value of 0.7 is considered the criterion for accepting that the construct has internal consistency. SmartPLS also allows for measuring the composite reliability and is a better measure of internal consistency as it is measured in relation to the total scale score variance (Hair *et al.*, 2014). Our model considered composite reliability values >0.7 to have good internal consistency (Hair *et al.*, 2011). All the alpha and composite reliability values of all the items in Table 5 meet the recommended criteria, indicating the establishment of internal reliability.

4.1.2 Validity testing. Validity testing was carried out to assess whether the items of the scale measure as it is intended to. Convergent and discriminant validity are the two measures considered for validity testing. Construct or convergent validity measures the correlation between the items of the studied constructs and determines how well the items describe the measured construct (Neuman, 2007). The average variance extracted (AVE) is used for

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Table 5. Measurement model analysis

Construct	Items	Factor loadings	Cronbach's alpha	Composite reliability	AVE
AC	AC1	0.837	0.79	0.867	0.628
	AC2	0.883			
	AC3	0.867			
	AC4	0.528			
AT	AT1	0.656	0.848	0.891	0.622
	AT2	0.844			
	AT3	0.775			
	AT4	0.822			
	AT5	0.83			
CM	CM1	0.831	0.737	0.85	0.655
	CM2	0.764			
	CM3	0.83			
EL	EL1	0.774	0.773	0.869	0.689
	EL2	0.875			
	EL3	0.838			
IN	IN1	0.598	0.818	0.874	0.705
	IN2	0.933			
	IN3	0.942			
PB	PB1	0.693	0.696	0.814	0.522
	PB2	0.704			
	PB3	0.741			
	PB4	0.751			
PR	PR1	0.916	0.946	0.963	0.896
	PR2	0.955			
	PR3	0.967			
RI	RI1	0.87	0.785	0.875	0.7
	RI2	0.859			
	RI3	0.778			
RL	RL1	0.904	0.924	0.952	0.869
	RL2	0.95			
	RL3	0.942			
SN	SN1	0.802	0.727	0.829	0.55
	SN2	0.825			
	SN3	0.634			
	SN4	0.69			

Source(s): Authors' own work

convergent validity, which indicates the average variance between the items and the corresponding constructs. AVE values ≥0.5 with a composite reliability ≥0.7 indicate a reliable measure of convergent validity (Hair *et al.*, 2014, 2021). Table 6 shows that the AVE and the composite reliability values are above 0.5 and 0.7, respectively, indicating that convergent validity is achieved.

Discriminant validity measures the correlation between the studied constructs to determine the degree to which the constructs differ from each other (Hair *et al.*, 2014). The squared root AVE values and the Heterotriat-Monotrait (HTMT) ratio are examined to determine discriminant validity. The squared root AVE values of the constructs are calculated and extracted from the Fornell-Larcker criterion analysis. To achieve discriminant validity, the AVE's square root values must be greater than the construct's internal correlation (Fornell and Larcker, 1981; Hair *et al.*, 2014). In recent years, some researchers have suggested that the HTMT ratio better represents the discriminant validity when compared to the Fornell–Larcker criterion. The HTMT ratio of <0.9 is considered for establishing discriminant validity

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Table 6. Discriminant validity analysis

Fornell-larcker criterion	AT	AC	CM	EL	IN	PB	PR	RI	RL	SN
AT	0.79									
AC	0.50	0.79								
CM	0.49	0.57	0.81							
EL	0.44	0.72	0.58	0.83						
IN	−0.16	−0.27	−0.28	−0.21	0.84					
PB	0.31	0.35	0.35	0.39	−0.14	0.72				
PR	−0.12	−0.17	−0.15	−0.14	0.32	0.04	0.95			
RI	0.56	0.49	0.59	0.39	−0.24	0.25	−0.04	0.84		
RL	0.02	0.19	0.17	0.24	−0.125	0.58	0.10	0.17	0.92	
SN	0.49	0.3	0.31	0.2	−0.10	0.28	−0.03	0.49	0.21	0.74
Hetrotrait-Monotrait ratio										
AT										
AC	0.60									
CM	0.61	0.75								
EL	0.53	0.89	0.78							
IN	0.16	0.32	0.30	0.21						
PB	0.40	0.48	0.49	0.53	0.16					
PR	0.13	0.19	0.18	0.17	0.39	0.07				
RI	0.66	0.62	0.77	0.50	0.25	0.35	0.05			
RL	0.06	0.22	0.20	0.29	0.16	0.71	0.12	0.20		
SN	0.59	0.41	0.41	0.28	0.17	0.41	0.09	0.64	0.30	

Note(s): The diagonal values in the Fornell–Larcker criterion matrix are the AVE's square root values
Source(s): Authors' own work

(Henseler *et al.*, 2015). The values in Table 6 indicate that AVE's square root values are higher than the internal correlation for all the constructs, and the HTMT values are <0.9.

4.2 Structural model analysis

After analyzing the outer model for reliability and validity, the inner model was examined to assess the predictive power of the endogenous constructs and to test the hypothesized relationships (Hair *et al.*, 2011). For testing the hypotheses, bootstrapping is carried out with 10,000 subsamples to provide reliable estimates of path significance.

4.2.1 Model predictive power. In the inner model analysis, the quality of the model was evaluated based on two important standards: the coefficient of determination (R^2) and cross-validated redundancy (Q^2). The coefficient of determination measures predictive accuracy and indicates the overall impact of studied constructs on the dependent variable (Hair *et al.*, 2014). The R^2 value of return intention in Table 7 indicates an overall moderate predictability of 0.56 (Sanchez, 2013).

The cross-validated redundancy measures the predictive relevance, and the value should be more than zero for the model to be relevant (Hair *et al.*, 2014). All the Q^2 values in Table 7 are greater than zero, indicating model fitness. Collectively, these findings validate that the model demonstrates both statistical accuracy and practical applicability in determining e-waste return intention.

4.2.2 Path analysis. Path analysis helps to examine the relations between studied constructs and assess their statistical significance. The path coefficients and T-values obtained from the analysis are used to determine the significance (Table 8).

The VIF was investigated to assess multicollinearity between the predictors. VIF values above 5 indicate collinearity problems (Hair *et al.*, 2016). The VIF values of all the constructs

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Table 7. Predictive relevance

Construct	*R*-square	Q^2 predict
Attitude	0.203	0.177
Perceived Behavioral Control	0.336	0.328
Return Intention	0.56	0.428

Source(s): Authors' own work

Table 8. Summary of path analysis

Hypothesis	VIF	Path coefficient	*t*-statistics	*F* square	Remarks
AT → RI	1.848	0.263***	4.834	0.089	Supported
SN → RI	1.393	0.263***	4.642	0.111	Supported
PB → RI	1.365	−0.052	1.219	0.006	Not supported
RL → AT	1.064	−0.098*	1.764	0.011	Supported
RL → PB	1	0.582***	14.78	0.507	Supported
EL → AT	1.064	0.467***	6.322	0.253	Supported
CM → RI	1.761	0.289***	4.726	0.108	Supported
AC → RI	1.849	0.054	0.919	0.004	Not supported
PR → RI	1.189	0.097*	1.999	0.021	Supported
IN → RI	1.23	−0.096*	2.106	0.019	Supported
PR × AT → RI	1.848	−0.138**	2.725		Supported
PR × PB → RI	1.91	−0.005	0.196		Not supported
PR × AC → RI	2.048	0.147**	2.638		Supported
IN × PB → RI	1.716	0.079*	1.99		Supported

Note(s): *$p < 0.1$; **$p < 0.01$; and ***$p < 0.001$

Source(s): Authors' own work

(Table 8) are lower than 5, establishing the fact that there are no collinearity issues. Table 8 also shows the variable's effect size (F-square). It is found that reverse logistics awareness (0.507) has the largest effect on return intention.

The result of the hypothesis testing is shown in Table 8. Table 8 shows that the effects of attitude ($\beta = 0.263$, $t = 4.834$), subjective norm ($\beta = 0.263$, $t = 4.642$), collection model ($\beta = 0.289$, $t = 4.726$), perceived risk ($\beta = 0.097$, $t = 1.999$) and incentive ($\beta = -0.096$, $t = 2.106$) on return intention are statistically significant, thus supporting H1, H2, H7, H9 and H13. The impact of reverse logistics awareness on attitude toward recycling ($\beta = -0.098$, $t = 1.764$) and PB control ($\beta = 0.582$, $t = 14.78$) and the impact of eco-literacy on attitude ($\beta = 0.467$, $t = 6.322$) were also statistically significant, thus supporting H4, H5 and H6.

4.2.3 Moderation analysis. The moderation analysis was conducted following the method suggested by Hair *et al.* (2016). From analysis (Table 8), it is observed that perceived risk negatively affects the relationship between attitude and return intention (β = -0.138, $t = 2.725$), supporting H10. However, perceived risk has a positive impact on the relationship between awareness consequences and return intention ($\beta = 0.147$, $t = 2.638$), supporting H13. Furthermore, incentive had a positive moderating effect on the relationship between PB control and return intention ($\beta = 0.079$, $t = 1.99$), thereby supporting H14.

5. Discussion

The empirical findings offer a theoretical contribution and contextual relevance in explaining consumer e-waste return behavior within an extended TPB framework. This current study

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enhances the body of literature on sustainable consumer behavior through empirical validation of a multi-construct model integrating logistical, cognitive, normative and risk factors – thereby offering an integrated view of the intention-forming processes for a developing economy.

5.1 Theoretical contribution

This research contributes theoretically by building on the TPB with six context-specific factors – reverse logistics awareness, eco-literacy, consequences awareness, collection mechanisms, perceived risk and incentives – and offers a multi-faceted understanding of consumer e-waste return behavior. It shifts the emphasis from remanufactured product adoption or generic recycling behaviors to the relatively unexamined area of return intentions for EoL and EoU electronic devices. By confirming the roles of informational, logistical and psychological factors, the study enhances TPB's predictive power in the sustainable consumption context. In addition, the study introduces moderation effects – perceived risk and incentive's influence on TPB pathways – providing a new theoretical perspective. This enriched framework provides the basis for future research in new markets.

First, this study empirically establishes that e-waste recycling attitude positively impacts the return intention of used electronic products, supporting previous literature related to environmental behavior. It indicates that individuals who believe that e-waste recycling is socially responsible and environmentally beneficial are more likely to recycle. This finding from the analysis is in line with previous studies on the impact of consumer attitudes toward purchase intention of remanufactured products (Khor and Hazen, 2017), reusing old products (Khan *et al.*, 2019) and intention to recycle (Yahya *et al.*, 2021). In India, where environmental issues are increasing as a result of urbanization and digitalization, this supports the significance of consumer's attitude in promoting sustainable behavior.

Second, the study emphasizes the importance of subjective norms, demonstrating the strong influence of social forces such as peer and parental aspirations in collectivistic societies. This is in line with Lee's (2009) study on green buying behavior and extends it to the e-waste scenario by confirming that societal support can greatly promote recycling participation.

The study also contributes to the theory by incorporating eco-literacy and reverse logistics awareness into the TPB model. These constructs, though less well-researched in TPB studies, emerged as significant motivators of attitude and PB control. Consumers who were aware of the environmental impacts of inappropriate disposal and had knowledge of return processes felt more confident and inclined to recycle e-waste, similar to findings by Ramayah *et al.* (2017) and Subramanian (2012). Consumers who were aware of the environmental impacts of inappropriate disposal and had knowledge of return processes felt more confident and inclined to participate in e-waste recycling activities, which is in line with the findings in the literature (Yahya *et al.*, 2021; Ramayah *et al.*, 2017; Babaei *et al.*, 2015; Subramanian, 2012).

A novel contribution of this study is the empirical testing of perceived risk and incentives as moderators of the TPB model. Perceived risk, particularly in relation to data privacy, was found to weaken the attitude-e-waste return intention relationship, establishing its position as a psychological barrier (Wang *et al.*, 2016; Yin *et al.*, 2014; Hazen *et al.*, 2012). This is complemented by the consumer behavioral inclination of "e-waste hoarding," where consumers keep old electronic devices such as mobile phones or computers for fear of personal data misuse. It has been established in earlier studies (Bovea *et al.*, 2018; Yin *et al.*, 2014) that such kind of digital hoarding is prevalent mainly in emerging economies. Conversely, incentives strengthen return intentions, especially when coupled with perceived control. This dual function – both direct and indirectly, through PB control – has not been previously investigated in the e-waste literature.

Finally, the research emphasizes the need to glocalize behavioral theories. By applying TPB to the Indian context, this study underscores how local environmental, social and digital conditions may influence consumer behavior in CLSCs. The integration of operational

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logistics constructs with psychological constructs in this study closes the theoretical gap between consumer behavior theory and logistics theory, proposing new interdisciplinary theoretical paths.

5.2 Managerial implication

Based on the result, we infer multiple managerial implications for practitioners, policymakers and stakeholders to enhance consumer involvement in e-waste return. To translate the research findings into actionable strategies, it is essential to build a systematic framework that links determinants of behavior to stakeholder-led interventions.

The study emphasizes the need for education programs for developing eco-literacy and awareness about reverse logistics. Public campaigns for creating awareness about the environmental impact of improper disposal of e-waste must be situation-specific and communicated through mass media, social media and schools. Such campaigns will encourage consumers to become more aware of the need for recycling and make them participate in the process. Government bodies, NGOs, educators, media outlets and other key stakeholders should collaborate to raise awareness while incorporating monitoring systems to track knowledge, attitude and return behavior changes in the long term.

Since perceived risk plays a significant role, especially with regard to data privacy, the research suggests that consumer assurance mechanisms are important. Governments will have to make sure that rules on certified data-wiping services are implemented, and firms and recyclers must send strong signals of privacy protection to boost customer trust and participation in return schemes, with retailers playing an important role in providing these services.

The study also highlights the need for the development of an easily accessible and convenient collection infrastructure. Multi-channel return mechanisms such as pick-up facilities, drop-off kiosks and retail partnerships must be scaled up, especially in regions with limited formal e-waste collection channels. This will help solve logistical challenges, particularly in low-awareness developing nations with limited knowledge of formal recycling channels. In this regard, local governments, producer responsibility organizations (PROs), logistics companies and retailers need to invest in scalable and low-cost collection facilities. Consumer return policies must ensure that easy e-waste return mechanisms are made part of the consumer experience, preferably through brick-and-mortar retail outlets or web-based return services.

Incentives were identified to have a multi-dimensional effect on return intentions and are considered a key component of the intervention strategy. Incentive structures must be contextual and clear. Rather than nonspecific monetary rewards, firms may provide in-kind incentives, e.g., warranties, loyalty rewards or discount prices on next purchases, aligning to consumer goals and motivations. Policy makers will also have to take into account integrated policy approaches that are not limited to Extended Producer Responsibility (EPR) but also to facilitate public-private partnerships. These strategies can facilitate e-waste take-back schemes, increase the number of collection facilities and improve monitoring of compliance processes, with decreased environmental impact and enhanced CLSC effectiveness. Retailers could also provide such incentives for returns, including loyalty programs or purchase vouchers, to encourage greater participation and complement EPR frameworks through strategic partnerships.

Since there is heterogeneity of consumer motivation, companies are encouraged to employ behavioral segmentation techniques to develop customized messages and return programs. With each consumer segment solving its own problem (e.g. privacy-conscious versus value-conscious recyclers), companies can enhance participation rates as well as maximize program effectiveness.

Lastly, producers must incorporate Design for Return features in their products, including disassembly ease, data erasure that is tamper-proof and readable return QR codes. Such design

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attributes will minimize psychological and operational resistance, ease return and enhance consumers' feelings of their control over the action. Moreover, digital nudges, like instantaneous environmental feedback, for instance or gamification (e.g. points for every return), can also diminish inertia and enhance participation rates for e-waste recycling.

6. Conclusion

In recent years, concerns regarding environmental pollution have been increasing due to a large number of electronic waste caused by the rapid growth in technology and the economy. With the increased environmental awareness, there is an increasing need to adopt ways to dispose of the EoU and EoL products to protect the environment's future. Recognizing this, researchers are becoming increasingly concerned with e-waste recycling and CLSC models. To this end, this study empirically investigates the CLSC to further develop its existing body of knowledge. This study extends the TPB for examining the psychological and contextual determinants of consumer return intention for electronic waste in India. It integrates constructs such as eco-literacy, reverse logistics awareness, perceived risk, incentives and collection methods to provide a better understanding of how consumers return e-waste in CLSC.

The findings of the study support the predictive validity of TPB's original constructs – attitude, subjective norms and PB control – and identify the pivotal role of informational and operational constructs. Notably, the findings indicate that eco-literacy and logistics consciousness enhance consumers' sense of efficacy and environmental stewardship, which in turn facilitates behavioral intention. Moderation analysis adds to the theoretical contribution, demonstrating that while perceived risk can suppress environmentally beneficial attitudes, it also enhances the impact of consequence awareness in generating behavioral intention under certain circumstances. Incentives, although small in magnitude, facilitate the translation of PB control into action, particularly when return logistics are viewed as feasible.

Collectively, this research helps to advance behavioral research on sustainable consumption by examining the contextual factors that affect TPB and demonstrating empirical evidence within a highly relevant but under-theorized environment.

6.1 Study limitations and future research directions

Despite the contributions, this research has some limitations. First, the adoption of a self-reported survey presents possible CMB and social desirability effects, even though procedural and statistical controls were implemented to minimize their effects.

Second, although the pilot survey improved the instrument's reliability and clarity, the pilot sample of doctoral students does not entirely represent the characteristics of the general population of consumers, particularly those residing in rural or low-income segments. Furthermore, urban bias can also exist, considering that most responses were gathered from metropolitan or semi-urban areas.

Third, the moderation analysis of the study, while insightful, only takes into account two moderators – perceived risk and incentive – and thus does not explore other potential moderating factors (e.g., trust in recyclers, moral obligation, convenience).

Finally, the model's consideration is limited to the intention to return e-waste and not to actual behavior. The intention-behavior gap could mean that some of the variables influencing actual recycling activities are not adequately measured.

Future studies can consider these limitations and build upon this study in meaningful ways. Longitudinal or mixed-methods can help researchers better understand how behavioral intentions regarding e-waste return can change and manifest as actual behavior over-time. Comparative analysis across different regions or nations can be conducted to assess the generalizability of the extended TPB framework, especially between developed and developing economies with diverse regulatory and infrastructural maturity. More theoretical

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depth can be derived by introducing other constructs, such as moral responsibility, environmental identity or collection agency trustworthiness, that can give further insight into the motivational and emotional aspects of sustainable behavior. Studies must explore segmented interventions across specific demographic or psychographic segments and thereby acknowledging consumer value, access and literacy heterogeneity. Finally, field experiments can evaluate the effectiveness of well-focused interventions – e.g. financial rewards, greater convenience or behavioral cues – on observed e-waste return behavior, filling the intention-action gap in real-life situations.

About the authors

Imnatila Pongen is Assistant Professor in the Department of Operations and Decision Sciences at T A Pai Management Institute, Manipal, India. She is also a research scholar specializing in Operations Management at the Indian Institute of Management Ranchi. Her scholarly work has been published in reputable journals such as the Journal of Cleaner Production (JCP) and Benchmarking. Her research interests encompass supply chain optimization, reverse channel coordination, green manufacturing and sustainable CLSC.

Dr Pritee Ray is Assistant Professor in the area of Operations Management at the Indian Institute of Management Ranchi. She obtained her PhD degree from IIT Kharagpur in the area of supply chain risk management. She has published her research work in top-tier International Journals such as EJOR, ANOR, etc. and won best research paper awards from several International Conferences. She also acts as an *ad hoc* reviewer in many International Journals and Conferences. She worked in academia for more than ten years. Her research interests include supply chain risk management, sustainable and closed-loop supply chain, circular economy, humanitarian supply chain and operations research.

Dr T.S. Vijay is Assistant Professor in the area of Marketing Management at the Indian Institute of Management Ranchi. He has completed his Fellow Program (PhD) in Marketing from the Indian Institute of Management Raipur. He has published his research work in JSM, JRCS, etc. He has more than ten years of academic and research experience. His research interest includes – E-WOM, online consumer behavior, online consumer reviews, celebrity endorsement, mobile apps and impulse buying behavior.

Dr Kannan Govindan is Professor in the area of Operations Management at Adelaide Business School, South Australia. He has published his research in many reputed journals over the years. He also acts in various capacities for many international journals: EIC – Cleaner Logistics and Supply Chain, Associate Editor – Transportation Research Part E: Logistics and Transportation Review and Executive Editor – *Journal of Cleaner Production*. He is also the Director and CSORSC at the University of Adelaide.

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Appendix

Table A1. Construct items

Constructs	Statements
Attitude	It is a good idea to return e-waste
	Recycling e-waste will make me feel satisfied
	It is beneficial to recycle e-waste
	Recycling e-waste is a shared responsibility
	I have a willingness to participate in recycling e-waste
Subjective norm	It is important what people important to me think about recycling e-waste
	People important to me would approve of me recycling e-waste
	People important to me expects me to return e-waste
	I will participate if people important to me return e-waste
Perceived behavioural control	I have plenty of opportunities to recycle e-waste
	I am aware of e-waste recycling
	I have sufficient time to recycle e-waste
	I have the ability to participate in e-waste recycling
Reverse logistics awareness	I know about companies taking back e-waste
	I know how to return e-waste
	I am aware of where to give the e-waste
Eco-literacy	Recycling e-waste is a rational use of used products
	Recycling e-waste can reduce environmental cost
	Improper disposal of e-waste causes environmental pollution
Collection method	An acceptable recycling system will encourage me to recycle e-waste
	Availability of multichannel will make e-waste recycling easier
	One should choose formal channels to recycle e-waste
Awareness consequences	Recycling reduces resource waste
	Recycling reduces environmental pollution
	Recycling creates a better environment for future
	Recycling reduces the cost of production
Perceived risk	I worry that personal information will be illegally recovered from e-waste
	I am fearful of leaking privacy information by companies collecting e-waste
	I worry that personal information will be misused
Incentive	I don't participate in e-waste recycling because there is no monetary incentive
	I will participate if I get some monetary incentive for e-waste recycling
	I will be more willing to participate if I get more monetary incentive
Return intention	I intend to return the used product
	I intend to actively encourage people around me to participate in e-waste recycling
	In the future I intend to choose products that promise to be recycled

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Corresponding author
Imnatila Pongen can be contacted at: imnatila.pongen@manipal.edu

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